



Technical Assessment: Cabify Rookie Excellence Program

Goals

Develop a module or package to manage drivers' geographic locations. It should allow updating their status and position, as well as reserving the driver nearest to a specified location. The solution should have tests.

Challenge

The module will track the geographic location of each driver. For every driver, the system will maintain a mapping between the driver ID (an opaque string) and their location. How this information persists (or does not) is left to the candidate's discretion.

To track the drivers and utilize this information, the solution provides a set of functions providing the following operations:

- Update a driver's location. Given a driver ID and its coordinates, the system will persist this information for future use. If the driver is unknown, this operation will add them to the system. If the driver already exists, their current location will be updated.
- Stop driver tracking. Given a driver ID, the system will cease tracking that driver after this call.
- Find the closest driver. Given a set of coordinates, the system will identify the driver nearest to that location, returning the driver ID. Only drivers currently being tracked should be considered.

Please define the necessary data types to model the information and any auxiliary functions needed to implement the requested functionality.

Example

```
Update driver0 at 0,0
Update driver1 at 10,10
Update driver2 at 9,9
Update driver0 at 9,8
Stop tracking driver2
Find the closest driver to 9,9 returns driver0
```



Delivery

You are expected to submit a **code file along with a README explaining the solution and the project.**

You will have **72 hours** to complete the delivery from the moment the challenge is sent.
You can send your response via email.